Linear vs. Non-Linear Classifiers in Pulsar Detection

A Comparative Study on Imbalanced Astrophysical Data

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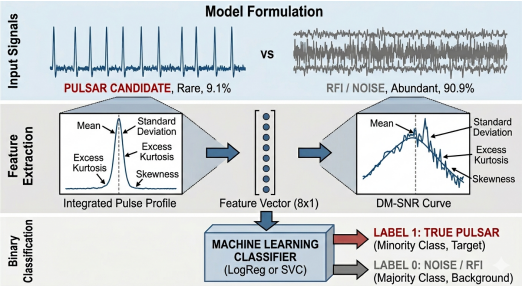
The Challenge: Pulsar Candidate Detection

The "Needle in a Haystack" Problem:

- HTRU2 dataset presents extreme class imbalance.
- **True Pulsars:** $\sim 9.1\%$ of the data.
- **RFI Noise:** $\sim 90.9\%$ of the data.

Objective:

- Build an automated, highly accurate classifier.
- Maintain transparency for astronomical verification (XAI).



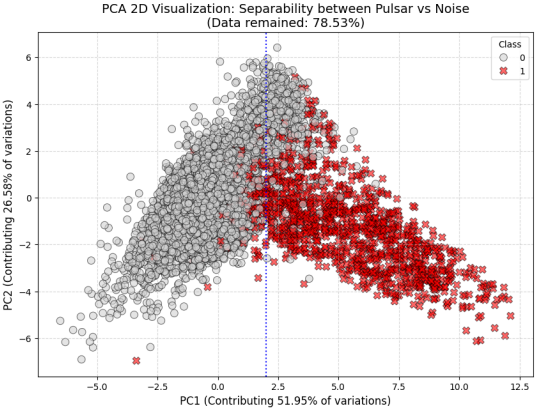
Data Engineering & Feature Separability

Data Preprocessing:

- 8 continuous statistical features.
- Applied **Z-score standardization** to stabilize distance-based margins.

Empirical Observation:

- 2D PCA visualizes strong structural separability.
- Indicates profound potential for linear classifiers.



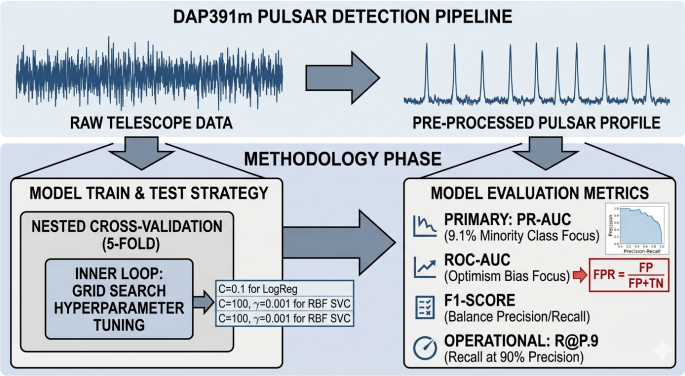
Methodology & Model Selection

Comparative Models:

1. **Logistic Regression:** Linear, transparent baseline.
2. **RBF SVC:** Complex, non-linear alternative.

Mitigating Imbalance:

- Implemented **Cost-sensitive learning**.
- Penalizes minority class misclassification heavily.



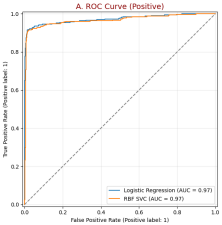
Evaluation Strategy: The Imbalance Paradox

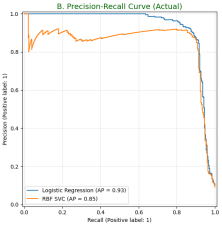
The Flaw of ROC-AUC:

- Massive True Negatives (noise) force False Positive Rate near zero.
- Creates dangerous **"Optimism bias"**.

The Primary Metric: PR-AUC

- Focuses strictly on minority class retrieval.
- **Operational Metric:** Recall at Target Precision 0.90 (R@P.9) simulates manual verification constraints.

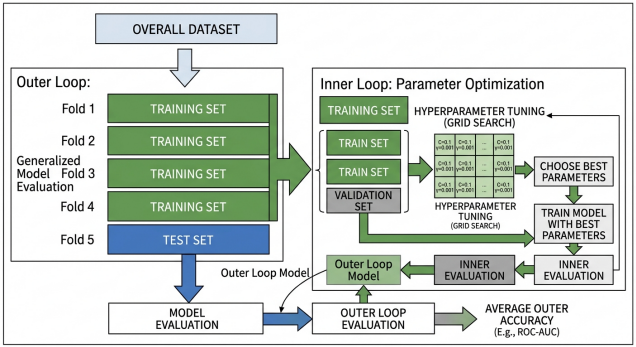




Experimental Setup

5-Fold Nested Cross-Validation:

- **Inner Loop:** Systematic Grid Search for hyperparameters ($C = 0.1$ for LogReg; $C = 100, \gamma = 0.001$ for SVC).
- **Outer Loop:** Unbiased evaluation on strictly unseen folds.
- **Stratification:** Perfectly preserves the 9.1% class distribution.



Experimental Results

Optimized Model Performance Comparison (Mean $\pm$ Std)

Model	PR-AUC	ROC-AUC	F1	R@P.9	Train Time (s)
LogReg	0.9226$\pm$.012	0.9768$\pm$.004	0.8406$\pm$.011	**0.8666$\pm$.039**	**0.21**
RBF SVC	**0.9268$\pm$.012***	**0.9788$\pm$.003**	**0.8680$\pm$.008**	0.8635$\pm$.040	24.28

* Difference with LogReg is not statistically significant ($p = 0.2905$, paired t-test).

- RBF SVC achieves a marginally higher global PR-AUC (0.9268).
- LogReg maintains a lead in high-purity operational Recall (R@P.9).
- **Efficiency:** LogReg is approximately **114x faster** to train.

Ablation Study: The Fragility of Non-Linearity

Default vs. Optimized Configuration:

- **LogReg (Robust):** Default PR-AUC was 0.9232. Post-tuning remained stable at 0.9226. The linear boundary is naturally optimal.
- **RBF SVC (Fragile):** Default PR-AUC was only 0.8739. Intensive Grid Search was mandatory to reach 0.9268 ($\approx +5.3\%$ gain).

Implication:

- LogReg offers "out-of-the-box" stability.
- Reinforces Occam's Razor: The dataset's inherent linearity renders complex configuration protocols redundant.

Configuration	PR-AUC	F1
LogReg (Default)	0.9232	0.8417
LogReg (Tuned)	0.9226	0.8406
SVC (Default)	0.8739	0.8618
SVC (Tuned)	**0.9268**	**0.8680**

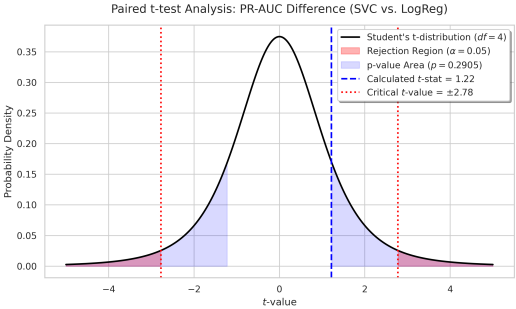
Discussion: Occam's Razor & Statistical Testing

Statistical Significance ($p = 0.2905$):

- Paired t-test confirms the PR-AUC difference is not statistically significant.
- 95% CI $[-0.0034, 0.0088]$ encompasses zero.

Application of Occam's Razor:

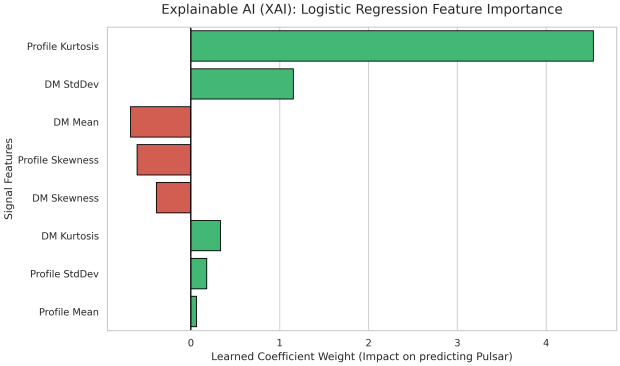
- LogReg provides equivalent predictive power with vastly lower latency.
- Complex non-linear mappings are computationally redundant for this dataset.



Bridging the Gap: Explainable AI (XAI)

HTRU2 Pulsar Diagnostic Agent:

- LogReg's linearity enables direct coefficient extraction ($w_j \cdot x_{ij}$).
- **Pulsar Signature (Green):** *Profile Kurtosis* (coef > 4.5) heavily dominates as the primary positive driver, perfectly identifying sharp, narrow pulsar peaks.
- **RFI Veto Filters (Red):** *DM Mean* and *Profile Skewness* act as strict negative drivers to filter out erratic, dispersed terrestrial noise.



Limitations and Deployment Risks

- **Feature Dependency (Garbage In, Garbage Out):** The model relies on hand-crafted upstream features. Flaws in initial signal folding will propagate as errors.
- **The RFI Subset Problem & Concept Drift:** Terrestrial interference (RFI) constantly evolves with technology. The model risks performance degradation when encountering novel, uncharacterized forms of artificial noise in future live surveys.

Conclusion

Key Takeaways:

1. **Sufficiency of Linearity:** Linear baselines are highly capable for capturing pulsar morphological features.
2. **Metric Integrity:** PR-AUC is mandatory; ROC-AUC is highly deceptive in extreme imbalance.
3. **Operational Supremacy:** Logistic Regression is the optimal framework (114x faster, out-of-the-box stability, equivalent accuracy).
4. **XAI Utility:** Model transparency transforms a black-box into a scientifically verifiable diagnostic tool.

Future Work & Strategic Directions

- **Ensemble Architectures (XGBoost/LightGBM):** Benchmarking state-of-the-art tree-based models to test the absolute limits of our linear sufficiency hypothesis.
- **Frequency-Domain Feature Engineering:** Moving beyond time-domain statistics by integrating Fast Fourier Transform (FFT) or Wavelets to resolve high-overlap "twilight zones".
- **Deep Learning (1D-CNNs):** Analyzing raw, unaggregated pulse profiles directly to eliminate dependency on hand-crafted features.
- **Unsupervised Anomaly Detection:** Deploying algorithms like Isolation Forests to discover "exotic" Fast Radio Bursts (FRBs) that defy known training distributions.

Thank You for Listening!